

urllib.error — Exception classes raised by urllib.request

Source code: <Lib/urllib/error.py>

The `urllib.error` module defines the exception classes for exceptions raised by [urllib.request](#). The base exception class is [URLError](#).

The following exceptions are raised by `urllib.error` as appropriate:

exception `urllib.error.URLError`

The handlers raise this exception (or derived exceptions) when they run into a problem. It is a subclass of [OSError](#).

reason

The reason for this error. It can be a message string or another exception instance.

Changed in version 3.3: `URLError` used to be a subtype of [IOError](#), which is now an alias of [OSError](#).

exception `urllib.error.HTTPError(url, code, msg, hdrs, fp)`

Though being an exception (a subclass of [URLError](#)), an `HTTPError` can also function as a non-exceptional file-like return value (the same thing that [urlopen\(\)](#) returns). This is useful when handling exotic HTTP errors, such as requests for authentication.

url

Contains the request URL. An alias for *filename* attribute.

code

An HTTP status code as defined in [RFC 2616](#). This numeric value corresponds to a value found in the dictionary of codes as found in [http.server.BaseHTTPRequestHandler.responses](#).

reason

This is usually a string explaining the reason for this error. An alias for *msg* attribute.

headers

The HTTP response headers for the HTTP request that caused the `HTTPError`. An alias for *hdrs* attribute.

Added in version 3.4.

fp

A file-like object where the HTTP error body can be read from.

exception urllib.error.**ContentTooShortError**(*msg*, *content*)

This exception is raised when the [urlretrieve\(\)](#) function detects that the amount of the downloaded data is less than the expected amount (given by the *Content-Length* header).

content

The downloaded (and supposedly truncated) data.